

Chapter 2: Linear Algebra: Vector Spaces & Transformations

2.1 Vector Spaces, Linear Combinations, and Span

Axiomatic Foundations

A vector space V over a field F is a set closed under vector addition and scalar multiplication satisfying the eight field axioms (associativity, commutativity, identity elements, and distributivity).

A linear combination of vectors $v_1, v_2, \dots, v_n$ is $c_1 \cdot v_1 + c_2 \cdot v_2 + \dots + c_n \cdot v_n$. The Span of a set of vectors is the collection of all possible linear combinations. If no vector in the set can be expressed as a linear combination of the remaining vectors, the set is Linearly Independent.

2.2 Basis and Dimension

Minimal Spanning Coordinate Systems

A Basis for a vector space V is a linearly independent set of vectors that spans V . The Dimension $\dim(V)$ is the number of vectors in any basis for V (an invariant property of the space).

2.3 Linear Transformations, Determinants, and Eigenvalues

Geometry of Matrix Operators

A mapping $T: V \rightarrow W$ is a Linear Transformation if $T(u + v) = T(u) + T(v)$ and $T(c \cdot v) = c \cdot T(v)$. Every linear map between finite-dimensional spaces can be represented as matrix multiplication $T(x) = A \cdot x$.

An eigenvector v of an $n \times n$ matrix A is a non-zero vector that satisfies $A \cdot v = \lambda \cdot v$, where scalar λ is the associated eigenvalue. Eigenvalues are found by solving the characteristic polynomial equation $\det(A - \lambda \cdot I) = 0$.

Key Takeaways:

- Rank-Nullity Theorem: For an $m \times n$ matrix A , $\text{Rank}(A) + \text{Nullity}(A) = n$.
- $\det(A) \neq 0$ if and only if matrix A is invertible.
- Eigenvectors represent invariant directions that only undergo scalar scaling under transformation A .